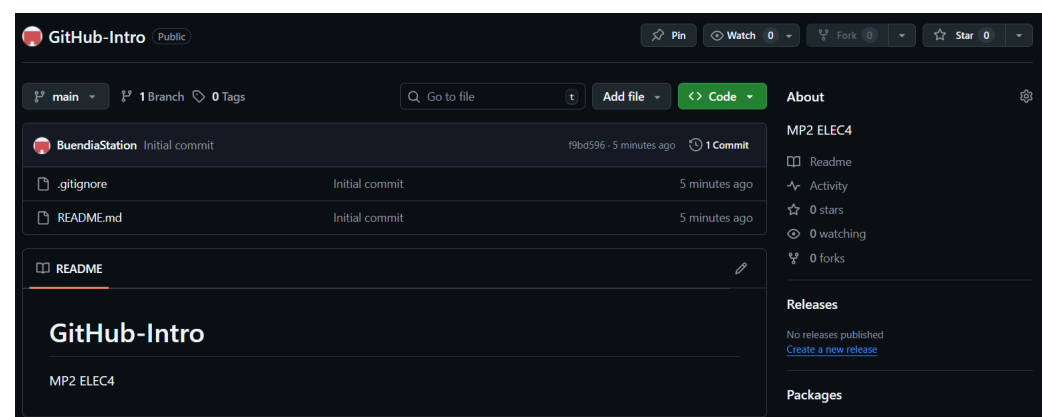


Problem Description:

- You are tasked with creating and managing a GitHub repository that will hold a simple Python program that prints "Hello, World!" You will simulate a full version control and collaboration workflow by completing the following steps:

Step 1: Setting Up Git and GitHub

- Create a GitHub repository:**
 - Name your repository GitHub-Intro.
 - Add a short description in the repository settings.
 - Initialize the repository with a README.md file and a .gitignore file (choose Python from the templates).
 - Take a screenshot of your repository's main page showing the README file.

Repository's main page showing the README file.

- Configure Git on your local machine:**
 - Use the command line to configure Git with your name and email, and take a screenshot of your terminal showing how you configure it.

Configured name and email in Git

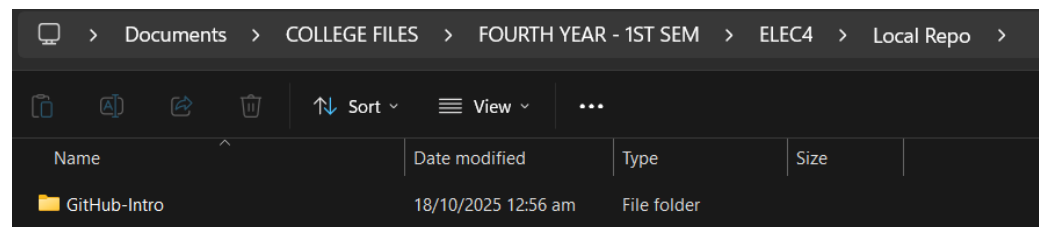
```
Pole@LAPTOP-SJGK48T3 MINGW64 ~  
$ git config --global user.name "Pole Andrei R. Buendia"  
  
Pole@LAPTOP-SJGK48T3 MINGW64 ~  
$ git config --global user.email "pbuendia.k12043566@umak.edu.ph"
```

Step 2: Cloning and Working with the Repository

- Clone the repository:**
 - Take a screenshot showing the git clone command and the local folder created.

git clone command

```
Pole@LAPTOP-SJGK48T3 MINGW64 ~  
$ cd "C:\Users\Pole\Documents\COLLEGE FILES\FOURTH YEAR - 1ST SEM\ELEC4\Local Repo"  
  
Pole@LAPTOP-SJGK48T3 MINGW64 ~/Documents/COLLEGE FILES/FOURTH YEAR - 1ST SEM/ELEC4/Local Repo  
$ git clone https://github.com/BuendiaStation/GitHub-Intro.git  
Cloning into 'GitHub-Intro'...  
remote: Enumerating objects: 4, done.  
remote: Counting objects: 100% (4/4), done.  
remote: Compressing objects: 100% (3/3), done.  
remote: Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)  
Receiving objects: 100% (4/4), done.
```

Local folder created

- **Add a Python file:**

- Inside the local repository folder, create a Python file named hello.py.
- Write a simple Python program that prints "Hello, World!" in the file.
- Save the file, and use the command line to add and commit the file:
- Take a screenshot of your terminal showing the git add and git commit commands.

git add and git commit commands

```
Pole@LAPTOP-SJGK48T3 MINGW64 ~/Documents/COLLEGE FILES/FOURTH YEAR - 1ST SEM/ELEC4/Local Repo/GitHub-Intro (main)  
$ git add hello.py  
  
Pole@LAPTOP-SJGK48T3 MINGW64 ~/Documents/COLLEGE FILES/FOURTH YEAR - 1ST SEM/ELEC4/Local Repo/GitHub-Intro (main)  
$ git commit -m "added hello.py"  
[main 9debd16] added hello.py  
1 file changed, 1 insertion(+)  
create mode 100644 hello.py
```

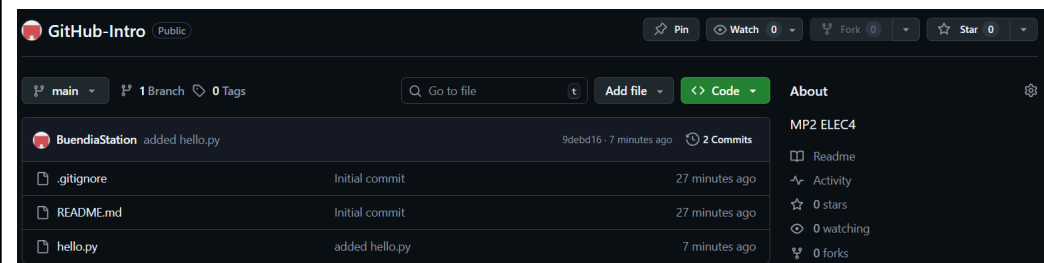
- **Push the changes to GitHub:**

- Push the changes from your local machine to your GitHub repository.
- Take a screenshot showing the successful push and the updated GitHub repository with the hello.py file.

Successful push

```
Pole@LAPTOP-SJGK48T3 MINGW64 ~/Documents/COLLEGE FILES/FOURTH YEAR - 1ST SEM/ELE
C4/Local Repo/GitHub-Intro (main)
$ git push origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 351 bytes | 351.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/BuendiaStation/GitHub-Intro.git
f9bd596..9debd16 main -> main
```

Updated GitHub repository with the hello.py file.



○

Step 3: Creating and Merging Branches

- **Create a new branch:**
 - Create a new branch called feature-branch for adding a new feature to the program.
 - In this branch, modify hello.py to take a user input and print their name.
 - Add and commit the changes.
 - Take a screenshot of the git branch, git add, and git commit commands.

git branch, git add, and git commit commands.

```
Pole@LAPTOP-SJGK48T3 MINGW64 ~/Documents/COLLEGE FILES/FOURTH YEAR - 1ST SEM/ELE
C4/Local Repo/GitHub-Intro (main)
$ git checkout -b feature-branch
Switched to a new branch 'feature-branch'

Pole@LAPTOP-SJGK48T3 MINGW64 ~/Documents/COLLEGE FILES/FOURTH YEAR - 1ST SEM/ELE
C4/Local Repo/GitHub-Intro (feature-branch)
$ git add .

Pole@LAPTOP-SJGK48T3 MINGW64 ~/Documents/COLLEGE FILES/FOURTH YEAR - 1ST SEM/ELE
C4/Local Repo/GitHub-Intro (feature-branch)
$ git commit -m "modify file to accept username input"
[feature-branch ff24827] modify file to accept username input
1 file changed, 4 insertions(+), 1 deletion(-)
```

- **Push the branch to GitHub:**
 - Push the feature-branch to GitHub.
 - Create a pull request on GitHub to merge feature-branch into main.

- Review the pull request, add comments if necessary, and merge it into the main branch.
- Take screenshots showing the pull request creation, review, and merge process.

Pull request creation

Open a pull request

Create a new pull request by comparing changes across two branches. If you need to, you can also [compare across forks](#). [Learn more about diff co](#)

 base: **main** < compare: **feature-branch** > ✓ Able to merge. These branches can be automatically merged.

 Add a title

Add a description

Write Preview

H B I           

Add your description here...

 Markdown is supported

 Paste, drop, or click to add files

Create pull request >

Pull request review

modify file to accept username input #1

 Open BuendiaStation wants to merge 1 commit into `main` from `feature-branch` 

 Conversation 0  Commits 1  Checks 0  Files changed 1

**BuendiaStation** commented [now](#) Owner ...

No description provided.



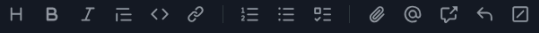
 modify file to accept username input ff24827

 **No conflicts with base branch**

Merging can be performed automatically.

Merge pull request  You can also merge this with the command line. [View command line instructions.](#)

**Add a comment**

Write Preview 

Add your comment here...

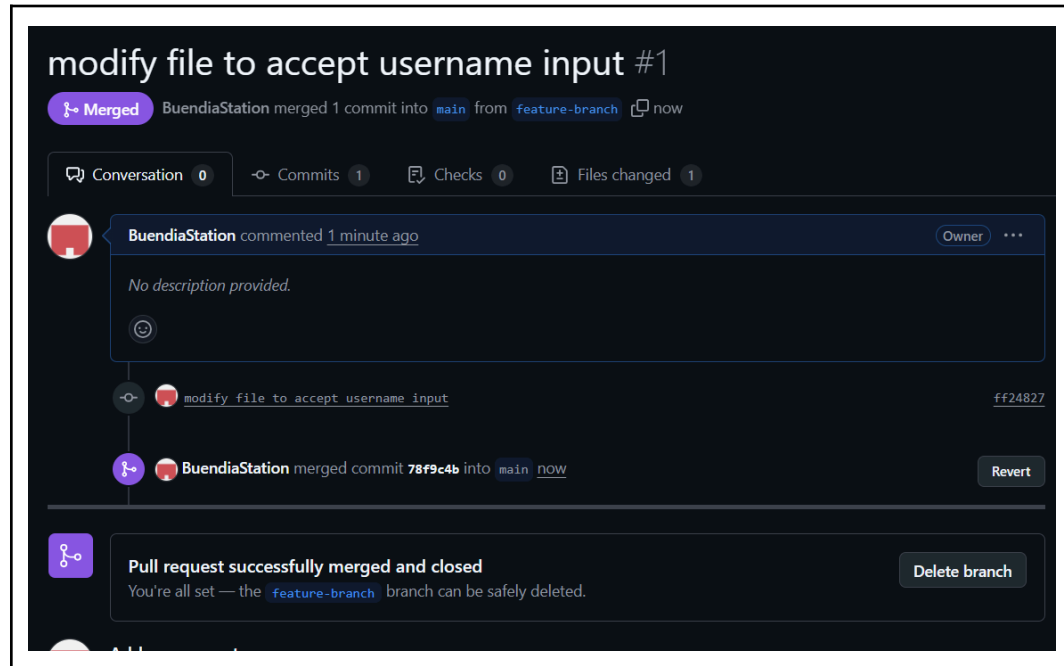
Pull request merge

 **No conflicts with base branch**

Merging can be performed automatically.

Merge pull request  You can also merge this with the command line. [View command line instructions.](#)

**Add a comment**

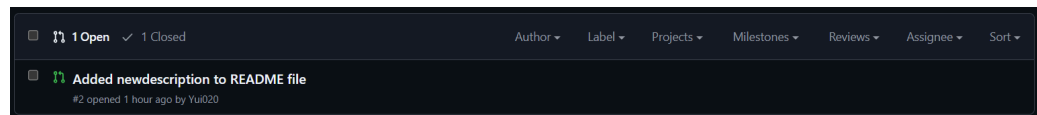


○

Step 4: Collaborating on GitHub

- Invite a collaborator:
 - Invite a classmate to collaborate on your repository using GitHub's "Collaborators" feature.
 - Ask your collaborator to clone the repository, create a new branch (fix-branch), and make a minor change to README.md (e.g., fixing a typo or adding a description).
 - Your collaborator should push their branch and create a pull request.
 - Review their pull request and merge it into the main branch.
 - Take a screenshot of the collaborator's pull request and the merge process.

Pull request



Pull request merge

Added newdescription to README file #2

 Open Yui020 wants to merge 1 commit into `main` from `fix-branch` 

 Conversation **0**  Commits **1**  Checks **0**  Files changed **1**



Yui020 commented 1 hour ago Collaborator 

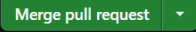
No description provided.



 Added newdescription to README file 2ba5876



 **No conflicts with base branch**
Merging can be performed automatically.

 Merge pull request  You can also merge this with the command line. [View command line instructions.](#)



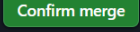
Commit message

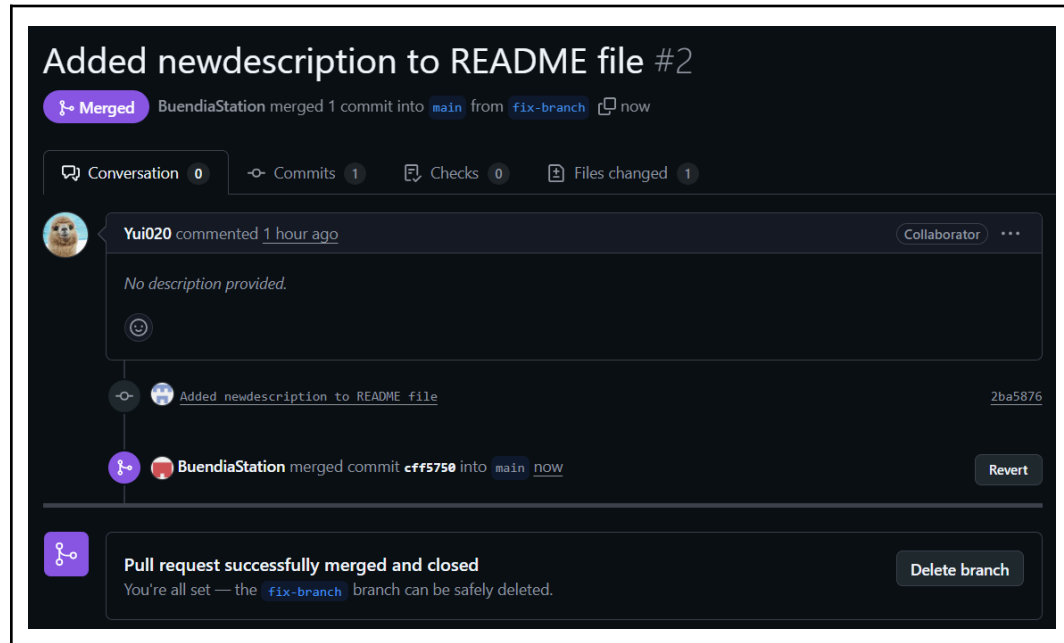
Merge pull request #2 from BuendiaStation/fix-branch

Extended description

Added newdescription to README file

This commit will be authored by 117883435+BuendiaStation@users.noreply.github.com.



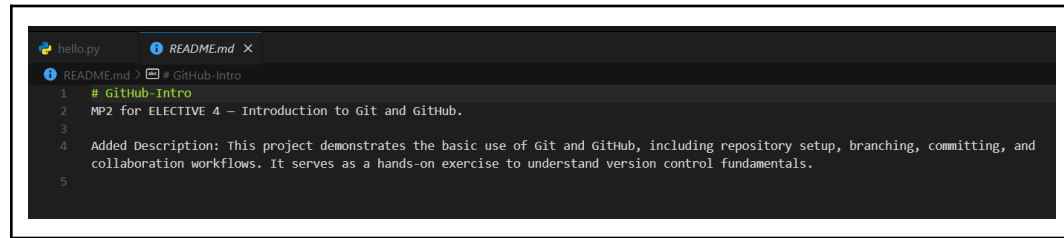
Step 5: Pulling Changes from GitHub

- Pull the latest changes:
 - After merging the collaborator's changes, use the git pull command to pull the latest changes to your local repository.
 - Take a screenshot showing the git pull command and the updated local repository.

Git pull

```
Pole@LAPTOP-SJGK48T3 MINGW64 ~/Documents/COLLEGE FILES/FOURTH YEAR - 1ST SEM/ELE
C4/Local Repo/GitHub-Intro (main)
$ git pull origin main
remote: Enumerating objects: 6, done.
remote: Counting objects: 100% (6/6), done.
remote: Compressing objects: 100% (4/4), done.
remote: Total 4 (delta 0), reused 3 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (4/4), 1.41 KiB | 76.00 KiB/s, done.
From https://github.com/BuendiaStation/GitHub-Intro
* branch          main       -> FETCH_HEAD
   78f9c4b..c9ff5750 main     -> origin/main
Updating 9debd16..c9ff5750
Fast-forward
 README.md | 4 +++-
 hello.py  | 5 ++++-
 2 files changed, 7 insertions(+), 2 deletions(-)
```

Updated local repo



The image shows a code editor window with two tabs: 'hello.py' and 'README.md'. The 'README.md' tab is active, displaying the following content:

```
1 # GitHub-Intro
2 MP2 for ELECTIVE 4 – Introduction to Git and GitHub.
3
4 Added Description: This project demonstrates the basic use of Git and GitHub, including repository setup, branching, committing, and
5 collaboration workflows. It serves as a hands-on exercise to understand version control fundamentals.
```